

Python

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\keywords{Machine Learning, Python}
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Item **7**

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[H]

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\node[box] (n13) at (10, 0) {}; \node[box] (n22) at (5, -3) {}; \node[box] (n21) at (0,
-3) {}; \node[box] (n23) at (10, -3) {}; \node[box] (n31) at (0, -6) {};
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(n23) -- (n22); \draw[arrow] (n22) -- (n21); \draw[arrow] (n21) -- (n31); \draw[arrow] (n31) --
(n32); \draw[arrow] (n32) -- (n33);

\caption{}
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1

MI

$$\hat{x}_{ij} = \bar{x}_j + \frac{\sum_{k=1}^n (x_{ik} - \bar{x}_k)(x_{kj} - \bar{x}_j)}{\sum_{k=1}^n (x_{ik} - \bar{x}_k)^2}$$
$$(x_{ij} - \bar{x}_i)$$

2

Isolation Forest

$$\text{ } = 2^{-\frac{\text{ }{c(n)}}$$

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}$$

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}$$

$$H(n) > 0.5$$

3

Z-score

$$x'_{ij} = \frac{x_{ij} - \bar{x_j}}{\sigma_j}$$

one-hot

4

$$Q_i = w_1 \cdot \text{ }_i + w_2 \cdot \text{ }_i + w_3 \cdot \text{ }_i$$

\$w_1=0.4, w_2=0.4, w_3=0.2\$

10000/ >99%

3

11/11/11

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[itemindent=2em] \item ██████████
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1. ■■■■■■

K-means

$$\min \sum_{i=1}^k \sum_{x \in C_i} |x - \mu_i|^2$$

$$k_4 C_i - i \mu_i$$

■■■■■■■■■■ [H] \centering \caption{■■■■■■}

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2. ■■■■■■

Collaborative Filtering Content-based

$$\text{■■■■} = \alpha \cdot \text{■■■■■} + (1-\alpha) \cdot \text{■■■■■■}$$

[illegible]

$$d(i,j) = \frac{\sum_{k=1}^m (x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j)}{\sqrt{\sum_{k=1}^m (x_{ik} - \bar{x}_i)^2} \sqrt{\sum_{k=1}^m (x_{jk} - \bar{x}_j)^2}}$$

[H] \centering \caption{■■■■■■■■}

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[illegible]

Item TOP-N

11/11/2019

[itemindent=2em] \item Coverage
 \item Diversity \item CTR \item
CVR

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[H] \centering \caption{■■■■■■■■■■}

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86.5%	&	+34.5%	\\	■ ■ ■ ■ ■ ■	&	48.2%	&	71.8%	&	+48.8%	\\
■ ■ ■ ■ ■ ■	&	32.1%	&	54.7%	&	+70.4%	\\	■ ■ ■ ■ ■ ■	&	68.5%	&
84.2%	&	+22.9%	\\								

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